

Objective function

Preconditioning

Michel Bierlaire

Optimization: principles and algorithms

Conditioning

The condition number is an indicator of the geometry of the function.

Its value influences the behavior of the algorithms

A good practice is to modify the problem formulation to obtain a better value of this indicator. This is called “preconditioning”. Some algorithms do that automatically.

Condition number of a matrix

Definition

$$\kappa(A) = \|A\| \|A^{-1}\|$$

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Symmetric positive semidefinite case

$$\kappa(A) = \lambda_1 / \lambda_n$$

Geometric interpretation

Consider

- ▶ $\nabla^2 f(x)$ positive definite,
- ▶ λ_1 its largest eigenvalue,
- ▶ d_1 a corresponding eigenvector.

$$Ad_1 = \lambda_1 d_1$$

$$d_1^T Ad_1 = \lambda_1 d_1^T d_1$$

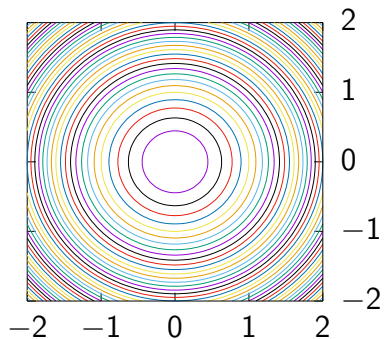
$$\lambda_1 = \frac{d_1^T Ad_1}{d_1^T d_1}$$

eigenvalue is the curvature along the eigenvector.

Condition number: ratio of largest to smallest curvature.

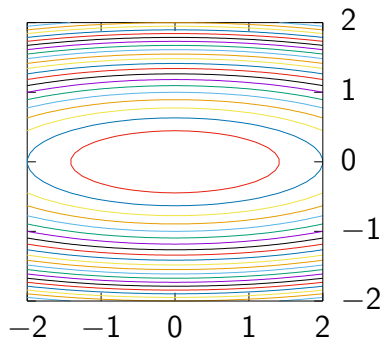
Well conditioned

$$f(x) = x_1^2 + x_2^2, \kappa(x) = 1$$



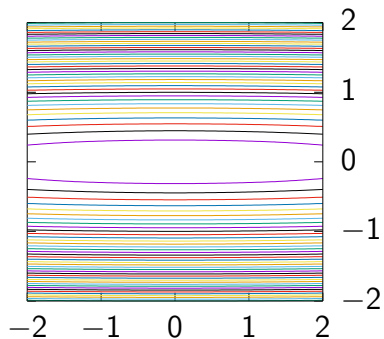
Ill-conditioned

$$f(x) = x_1^2 + 10x_2^2, \kappa(x) = 10$$



Ill-conditioned

$$f(x) = x_1^2 + 100x_2^2, \kappa(x) = 100$$



Change of variable

$$x' = Mx \text{ and } x = M^{-1}x'$$

$$\tilde{f}(x') = f(M^{-1}x')$$

$$\nabla \tilde{f}(x') = M^{-T} \nabla f(M^{-1}x')$$

$$\begin{aligned}\nabla^2 \tilde{f}(x') &= M^{-T} \nabla^2 f(M^{-1}x') M^{-1} \\ &= M^{-T} \nabla^2 f(x) M^{-1} .\end{aligned}$$

Preconditioning: Find the change of variable M such that the condition number of $\nabla^2 \tilde{f}(x')$ is better than $\nabla^2 f(x)$

Summary

The condition number reflects the geometry of the function through the ratio of extreme curvatures.

It can be changed using a change of variables. It is called preconditioning.